Combinatorial Lane Merging Problems

Experimental Math Seminar 10/15

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The Problem

A two lane road approaches a traffic light. Shortly after the intersection, the left lane merges into the right lane.

Car preferences: right (1) or shortest lane (2). (*Tie?* $\implies$ *Right lane*)

The list of car preferences form an *arrival sequence*, denoted $A \in \{1, 2\}^n$.

Example: $A = 11222112$.

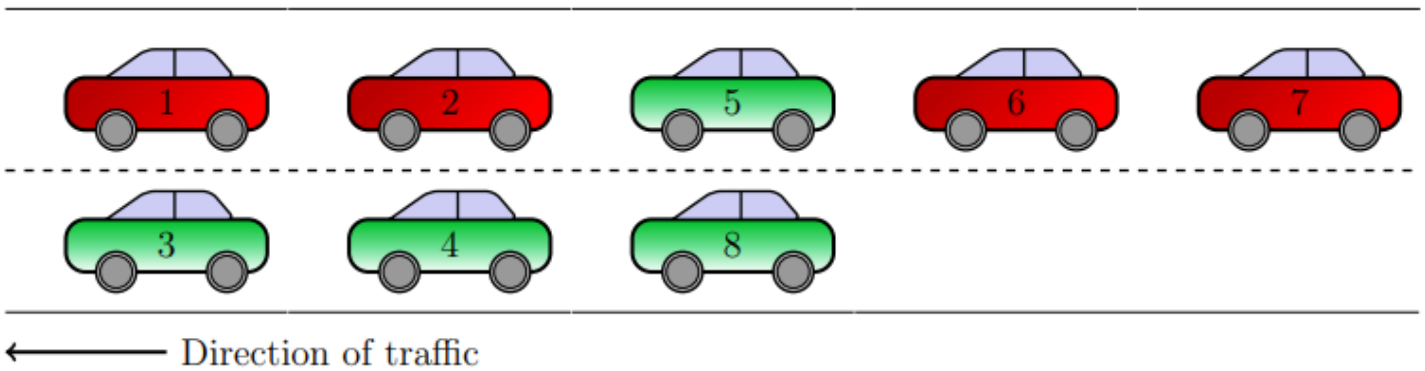


Image credit Bardenova et al. (2023)

The Problem (cont.)

Motivating Question: How many arrival sequences in $\{1,2\}^n$ end with m cars in the right lane and l in the left? *Notationally:* $F(m,l) = ???$

Observation: The left lane is *always* at most as long as the right lane.
★ Visualize with a Dyck path!

Def. A **Dyck path** is an up-right path which starts at (0,0) and never crosses below the diagonal.

Define a path from an arrival seq.:

- Car in right lane $\mapsto$ Up edge
- Car in left lane $\mapsto$ Right edge

Example: $A = 11222112$

Another: $A' = 11221112$
★ Not a bijection!

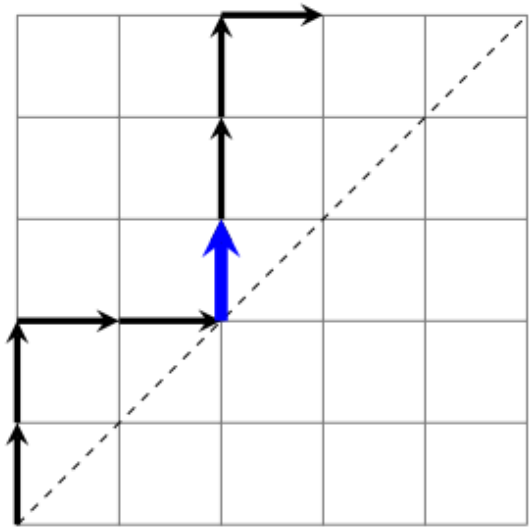


Image credit Bardenova et al. (2023)

Two-Lane Results

Dyck paths help us develop a recurrence:

$$\begin{aligned}
F(m,l) &= F(m,l-1) + F(m-1,l) && m > l+1 \\
F(m,l) &= F(m,l-1) + 2F(m-1,l) && m = l+1 \\
F(m,l) &= F(m,l-1) && m = l \\
F(0,0) &= 1 \\
F(m,0) &= 2
\end{aligned}$$

And a closed form:

$$F(m,l) = 2\binom{m+l}{m} \qquad m > l$$

$$F(m,l) = \binom{2m}{m} \qquad m = l$$

Can we do better?

Three-Lane Problem

What if we now have three lanes that all merge into the rightmost lane?

Car preferences for $A \in \{1, 2, 3\}^n$:

- (1) = Right lane
- (2) = Shortest of the middle or right lane
- (3) = Shortest of all lanes
  - Tiebreak opts for rightmost tied lane

Example: $A = 12131233$

FIGURE HERE

General Problem

Car preferences for $A \in \{1, 2, \ldots, k\}^n$:

- (i) = Shortest of the i rightmost lanes for any $1 \leq i \leq k$
- $*$ Tiebreak still opts for rightmost tied lane

Main Question: $F(x_1, x_2, \ldots, x_k) = ???$
★ x_i = length of i-th lane

To visualize our sequences and construct recurrences . . . Uh-oh.

- 3-dimensional Dyck paths? :/
- k-dimensional Dyck paths? :(

BUT we do still know that lane lengths are weakly decreasing from right to left! So . . .

Standard Young Tableaux

Def. A **SYT** is a left-justified array of n numbers such that:

- Each number in $\{1, 2, \ldots, n\}$ appears in exactly one position
- Row lengths are weakly decreasing from top to bottom
- Entries are strictly increasing from $L \to R$ and $T \to B$

Def. The **shape** of T is a list $[t_1, t_2, \ldots, t_k]$ of the row lengths from top to bottom.

Define a SYT T from an arrival seq. A:

- Car j in i-th lane $\implies$ entry j in i-th row of T

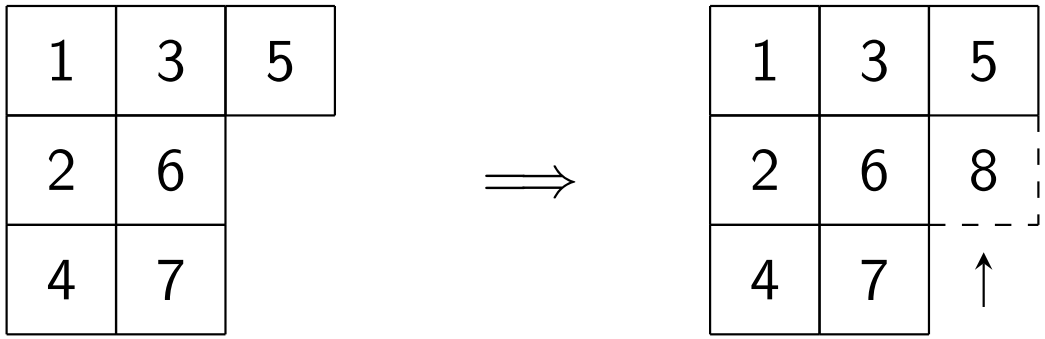
Example: $A = 1312223133$

1	3	5	8
2	4	6	
7	9	10	

Mapping to SYT

Question: How many A map to a given T? *Notationally:* $\#A(T)$.
Recall: Tied lanes $\implies$ both multiple prefs. produce same result!

Example: $A = 12131233$



Car 8 could have preference 2 or 3 and would still end up in middle lane!

Def: A **bounce** occurs when a car w/ pref. $i > 1$ tie breaks to the right of lane i.

Bounces

Lemma.

If T is a SYT and T' is the SYT obtained by deleting the largest element, n, from cell c of T, then

$$\frac{\#A(T)}{\#A(T')} = \beta(c)$$

where $\beta(c)$ is the size of bounce, or the number of open cells below c.

Proposition.

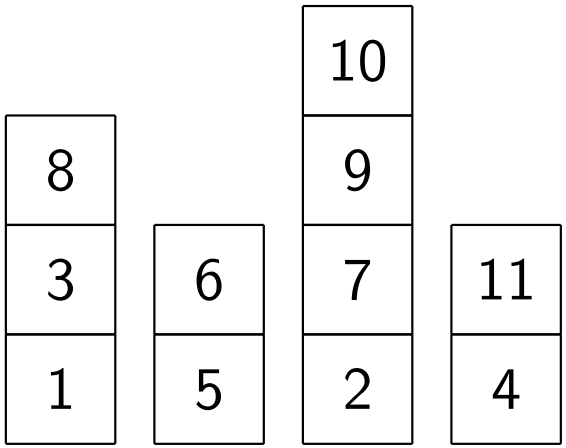
$$\#A(T) = \prod_{c \in T} \beta(c)$$

Note: First column has a special definition of $\beta(c)$. See Hive. & Zeil. for details.

Voting Sequences

In a voting sequence in $\{1, 2, \ldots, k\}^n$, each of n voters votes for one of k candidates. When finished, there are p_1 votes for the 1st place candidate, p_2 votes for the 2nd place candidate, etc.

Example: $V = 13142231334$



$p_1 = 4$ Candidate #3
$p_2 = 3$ Candidate #1
$p_3 = 2$ Candidate #2 (Tied)
$p_4 = 2$ Candidate #4 (Tied)

Final vote counts are weakly decreasing by popularity.
This feels familiar . . .

Voting Sequences (cont.)

We can also construct SYT for voting sequences!
Example: $V = 11122333$

1	2	3	9
4	5	8	
6	7		

Lemma.

$$\frac{\#V(T)}{\#V(T')} = \beta(c)$$

Here, $\beta(c)$ is the number of candidates with tied popularity that Voter n could vote for and produce the same final vote count.

Closed Form

Lemma.

Majority voting sequences are counted by the closed form

$$F(v_1, \ldots, v_k) = \binom{n}{v_1, \ldots, v_k} \cdot \binom{k}{p_1, \ldots, p_r}$$

where $p_1, p_2, \ldots, p_r$ is the partition of k defined by $(v_1, \ldots, v_k)$ in frequency notation.

Example of frequency notation for a partition:

$$[1, 1, 2, 2, 2, 2, 3] \to 1^2 2^4 3^1 \to (2, 4, 1)$$

Proof outline:

- n votes cast. Choose v_1 for Candidate 1, v_2 for Candidate 2, etc.
- k candidates total. Choose p_1 to be (tied) for 1st place, p_2 to be (tied) for 2nd place, etc.

Conclusion

Corollary.

Since both arrival sequences and voting sequences satisfy the same recurrence,

$$F(x_1, \ldots, x_k) = \binom{n}{x_1, \ldots, x_k} \cdot \binom{k}{p_1, \ldots, p_r}$$

Other comments:

- Agrees with Bardenova et al. closed form!
- Mapping $A \to V$ can be constructed using our recurrence
- Tables for summed lane lengths can be found in Hive. & Zeil.
- ★ For right lane, these sequences are in the OEIS!
  $\hookrightarrow$ A230137, A265083, A265084, A265085, etc.

Thanks & Questions

Selected Reading:

- V. Bardenova, E. Insko, K. Johnson, and S. Sullivan, A combinatorial model for lane merging, *Journal of Integer Sequences* **26** (2023) Article #23.5.3
- E. Insko, K. Johnson, and S. Sullivan, A combinatorial model for lane merging: Limited capacity lanes, *ECA* **6:2** (2026) Article #S2R9.
- A. Hiveley and D. Zeilberger, The Combinatorics of Multi-Lane Merging, arXiv, arxiv.org/pdf/2607.26334

Thanks! Questions?

Mapping is Hard

HAVE AN EXAMPLE OF SHITTY MAPPING PREPARED AS A BONUS SLIDE